VANISHING OF THE COKERNEL AND CONSTRUCTIBILITY OF THE KERNEL OF FROBENIUS-MINUS-ONE ON A VECTOR BUNDLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-etale.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Disclosure: two soft spots, both anchored or mechanical. (1) 'A computation shows that dx_i is zero in Omega' (proof line 4469): the differentiation of the displayed defining equations of A' is not written out, but is a direct mechanical differentiation of the displayed relations. (2) Faithful flatness: 'the reader can show that A -> A' is faithfully flat by verifying directly that all geometric fibres are nonempty, however this also follows from Lemma etale-cohomology-lemma-F-1' (proof lines 4477-4481) – the direct verification route is offered to the reader, but the claim is anchored to the named lemma. No verbatim 'we omit' occurs in the proof.

Allowed standard prerequisites (editorial).

- Frobenius minus one is surjective on the affine line (Etale Cohomology, Lemma etale-cohomology-lemma-F-1, canonical tag 0A3L). cited at proof lines 4477-4481 and the paragraph at statement lines 4447-4449. Anchors faithful flatness of A -> A' (all geometric fibres nonempty) and is the special case this lemma generalizes.
- Constructibility for etale morphisms (Etale Cohomology, Lemma etale-cohomology-lemma-jshriek-constructible, canonical tag 03S8). cited at proof lines 4491-4493. The terminal step for the kernel: a scheme affine etale over S has constructible sections sheaves.

Source reference. Original lemma 0A3L: sources/stacks-chow/tags.txt, source lines 12744–12744. Cited at proof lines 4477-4481: anchors faithful flatness of A -> A' (the lemma this one generalizes, per the paragraph at statement lines 4447-4449).

Source reference. Original lemma 03S8: sources/stacks-chow/tags.txt, source lines 4957–4957. Cited at proof lines 4491-4493: the kernel is represented by a scheme affine etale over S, so its sections sheaf is constructible.

Lemma 0.2. *Let p be a prime number. Let S be a scheme over $\mathbf{F}_p$. Let $\mathcal{E}$ be a finite locally free $\mathcal{O}_S$-module viewed as an $\mathcal{O}_S$-module on $S_{étale}$. Let $F : \mathcal{E} \to \mathcal{E}$ be*

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a homomorphism of abelian sheaves on $S_{étale}$ such that $F(ae) = a^p F(e)$ for local sections a, e of $\mathcal{O}_S$, $\mathcal{E}$ on $S_{étale}$. Then

$$\mathrm{Coker}(F - 1 : \mathcal{E} \to \mathcal{E})$$

is zero and

$$\mathrm{Ker}(F - 1 : \mathcal{E} \to \mathcal{E})$$

is a constructible abelian sheaf on $S_{étale}$.

Proof. We may assume $S = \mathrm{Spec}(A)$ where A is an $\mathbf{F}_p$-algebra and that $\mathcal{E}$ is the quasi-coherent module associated to the free A-module $Ae_1 \oplus \ldots \oplus Ae_n$. We write $F(e_i) = \sum a_{ij} e_j$.

Surjectivity of $F - 1$. It suffices to show that any element $\sum a_i e_i$, $a_i \in A$ is in the image of $F - 1$ after replacing A by a faithfully flat étale extension. Observe that

$$F(\sum x_i e_i) - \sum x_i e_i = \sum x_i^p a_{ij} e_j - \sum x_i e_i$$

Consider the A-algebra

$$A' = A[x_1, \ldots, x_n]/(a_i + x_i - \sum\nolimits_j a_{ji} x_j^p)$$

A computation shows that $\mathrm{d}x_i$ is zero in $\Omega_{A'/A}$ and hence $\Omega_{A'/A} = 0$. Since A' is of finite type over A, this implies that $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is unramified and hence is quasi-finite. Since A' is generated by n elements and cut out by n equations, we conclude that A' is a global relative complete intersection over A. Thus A' is flat over A and we conclude that $A \to A'$ is étale (as a flat and unramified ring map). Finally, the reader can show that $A \to A'$ is faithfully flat by verifying directly that all geometric fibres of $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ are nonempty, however this also follows from Étale Cohomology, Lemma 0A3L. Finally, the element $\sum x_i e_i \in A'e_1 \oplus \ldots \oplus A'e_n$ maps to $\sum a_i e_i$ by $F - 1$.

Constructibility of the kernel. The calculations above show that $\mathrm{Ker}(F - 1)$ is represented by the scheme

$$\mathrm{Spec}(A[x_1, \ldots, x_n]/(x_i - \sum\nolimits_j a_{ji} x_j^p))$$

over $S = \mathrm{Spec}(A)$. Since this is a scheme affine and étale over S we obtain the result from Étale Cohomology, Lemma 03S8. □